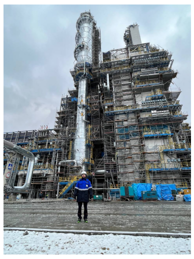
Denis Meleshkin

melesh.denis@gmail.com LinkedIn profile Portfolio

Curriculum vitae



Education

2023 – · · · · **M.Sc. Energy Engineering.**
University of Padua, Industrial Engineering department.
Studied such subjects as: Energy Systems, Advanced Control Systems, Combustion, Renewable Energy Technology, Wind and Hydraulic turbines and other.

2022 – 2023 **M.Sc. Cryogenic equipment and technologies.**
Bauman Moscow State Technical University, Power Engineering department
Studied for the first semester and passed all the exams with the highest grades. Then dropped out, as decided to continue education abroad.

2018 – 2022 **B.Sc. Refrigeration, Cryogenic Equipment And Life Support Systems.**
Bauman Moscow State Technical University, Power Engineering department
Thesis title: *Helium cryogenic cooling system with a capacity of 2 kW per 100-150 K.*
Diploma with honour.

Publications

Journal Articles

- **D. Meleshkin**, S. Pochatkov, and I. Svetlov, “Vibrodiagnostics method for determining damage to composite materials,” *Grand Mechanic*, no. 5, pp. 337–342, 2022. URL: `https://www.elibrary.ru/item.asp?id=48445134`.

Conference Proceedings

- D. Ershov, A. Mironov, **D. Meleshkin**, S. Pochatkov, and I. Svetlov, “Methods of numerical analysis and vibration diagnostics for assessing deformations in mechanical systems,” in *Students’ scientific spring (2022)*, Moscow, Russia, 2022, pp. 22–24. URL: `https://www.elibrary.ru/item.asp?id=49480593`.

Working Experience

2022 – 2023 **Jr. Design Engineer**. PJSC "Cryogenmash".
The main area of work was the calculation and modelling of cryogenic systems in Aspen HYSYS software, and rating of heat exchangers.

2021 – 2022 **Purchasing Manager**. LLC STATMED.
The main responsibilities included: market analysis, interaction with partners, organization of international purchases of various volumes.

Working Experience (continued)

Internships

2022 **Production practice at the Amur Gas Processing Plant.**
During the internship the operation of the natural gas purification, separation and liquefaction line, as well as the production of liquid helium, was studied.

2021 **Technological practice at the NPO Nauka enterprise.**
During the internship a technological process for manufacturing a pressure valve part was developed.

Skills

Engineering
- Design and analysis of energy systems and devices, modelling and simulation;
- Planning and caring out an experiment, processing the results;
- Entropy and exergy analysis;
- Life Cycle Assessment;
- Developing of the design documentation.

Software
- Calculations: MATLAB, PTC Mathcad, MS Excel, Julia;
- Modelling and simulations: Simulink and Simscape, ANSIS, OpenFOAM, Aspen HYSYS;
- Design and documentation: Autodesk Inventor, Autodesk AutoCAD, Solid-Edge, CFturbo;
- Reporting and publishing: LaTeX, MS Word, MS PowerPoint, MS Visio, GIT.

Language
Strong reading, writing and speaking competencies for English, proved by IELTS.

Misc.
Academic research, working in groups, management skills.
Leadership experience: during the studying for the Bachelor's degree, I was elected head of the study group, held this honorary position until graduation and was awarded a special payment for it.